Question	Answer	Marks	Guidance
4(a)	$a = -\frac{1}{5}$	**B1**	
	$b = \frac{7}{25}$	**B1**	
		2	
4(b)	$\mathrm{f}(x) = \begin{cases} \frac{1}{5} & 1 \leqslant x < 4, \\ \frac{1}{25}x & 4 \leqslant x \leqslant 6, \\ 0 & \text{otherwise.} \end{cases}$	**M1**	Attempt at differentiating both parts.
		A1	All correct, fully defined including domain. Domain must cover all reals, must not overlap at boundaries.
		2	
4(c)	$\mathrm{E}(X^2) = \frac{1}{5}\int_1^4 x^2 \mathrm{d}x + \frac{1}{25}\int_4^6 x^3 \mathrm{d}x = \left[\frac{1}{15}x^3\right]_1^4 + \left[\frac{1}{100}x^4\right]_4^6$	**M1**	Attempt to integrate $x^2\mathrm{f}(x)$ with correct limits. Can be implied by answer of $\frac{73}{5}$ or 14.6.
	$\mathrm{Var}(X) = \frac{73}{5} - \left(\frac{529}{150}\right)^2$	**M1**	Variance formula used with *their* $\mathrm{E}(X^2)$.
	2.16	**A1**	
		3	
4(d)	10th percentile: $\frac{1}{5}x - \frac{1}{5} = \frac{1}{10}$ 90th percentile: $\frac{1}{50}x^2 + \frac{7}{25} = \frac{9}{10}$	**M1**	Award M1 for a correct method to find at least one of the 10th or 90th percentile. FT *their* a and b.
	10th percentile = 1.5	**A1**	
	90th percentile = $\sqrt{31}$	**A1**	AWRT 5.57.
		3	